

FYS 191MCS26 — Bits, Bytes, and Modern Computing Systems

Topic Description

This First Year Seminar (FYS) contains a lot of the same supportive content as other FYS courses at UMass. Outside of that common content, this course asks one question: how does a machine that knows only two symbols — 0 and 1 — end up holding your photos, your music, your files, and a connection to a server on the other side of the planet? **No programming experience is assumed, and none is required.**

Learning Outcomes

- Students will gain awareness of the “hidden curriculum” of college
- Students will gain confidence with making connections across campus
- Students will practice maintaining a balance and attending to self-care
- Students will be able to explain how numbers, text, and images are represented in binary
- Students will be able to describe in plain language how memory, an operating system, and a network move data from one place to another

Logistics

Textbook: No textbook is required for this course. All slides and materials are posted on the course website and on Canvas.

Course Format: We meet once a week for 50 minutes. Sections A and B meet on Mondays; Sections C and D meet on Wednesdays. Times and room: see SPIRE / TBD. Each class follows the same shape — a short opening, a group activity, and then the explanation. Bring paper and a pen every week, and a laptop on the days I announce in advance.

The best way to reach me outside of class is via email at aupreti@umass.edu.

Assignments and Grading

All major assignments will be submitted via Canvas, and all feedback and grades will also be communicated via Canvas. I will cover the submission process in class. Feedback will not be posted prior to the due date, but students may submit an assignment as early as they wish.

Your success in this course is important. All First Year Seminar instructors are partnered with Student Success and your academic advisors to assist you in better understanding course material which can aid you on your path to success. Resources available to students include: Supplemental Instruction, ExSEL Group Tutoring, and 1:1 Tutoring. Visit the Learning Resource Center online at <http://www.umass.edu/lrc>.

Throughout the semester, your instructor will communicate with Student Success & academic advisors regarding your progress in the course. If you are contacted, please consider scheduling appointments such as tutoring or academic advising and talk with your professor. Referrals are not punitive and are meant to assist you in connecting with resources at UMass. Please email academicalert@umass.edu if you have any questions or need assistance connecting with resources.

Grading Breakdown

Engagement & Participation — 30%. Participation includes arriving to class on time and prepared for the day’s scheduled activities. Because most classes are built around a group activity, it also means working with your group and being willing to try an idea that might be wrong — in this course, a wrong first attempt is usually the point. It also includes classroom conduct. The expectation is that our class feels like a welcoming and safe learning space for everyone; to have “good” participation is to act in accordance with that expectation.

Canvas Check-ins — 10%. Short weekly questionnaire on Canvas. These should take about 3–4 minutes to complete.

Guest Lecture Reflections — 20%. There will be 4 guest lectures. Students will write a 350-word reflection for each. The top 2 scores will count toward the final grade (10% each).

College Homework — 30%

- *Event Assignment (10%):* Attend one campus/career/club/LRC event and write a short reflection.
- *Resources Scavenger Hunt (10%):* Complete a guided worksheet designed to demystify UMass support resources.
- *Study Report (10%):* Submit a study strategies reflection/report.

Final Show-and-Tell — 10%. In the last class, each student gives a 3-minute presentation on one everyday thing that is secretly made of bits — a file format, a device, an app, a piece of hardware — explaining what is going on underneath using ideas from the course. No slides required. A one-paragraph proposal is due on Canvas two weeks beforehand.

Course Policies

Attendance – Attendance and participation are expected components of the course. You are permitted 2 absences without penalty to your class participation grade and without requirement for notification or documentation to me. No distinction will be made between excused and unexcused absences except as governed by university policy. Generally, three late arrivals will be counted as an absence. Arriving to class more than 20 minutes late also counts as an absence.

Announcements – Canvas and email will be used for conveying important class announcements.

Accommodation Policy – Students with a documented disability are encouraged to communicate their needs to the instructor at their earliest convenience. If you aren't sure if you qualify for an accommodation, or if you have any questions about your right to accommodations, please contact the Office of Disability Services: <http://www.umass.edu/disability/students>

Cell Phones / Computers – Some class sessions use laptops for web resources and hands-on exercises, so bring your laptop on the days I announce. Most activities are done on paper in groups; during those, please keep phones and laptops away. I expect that you will use your laptop and cell phone responsibly during class, i.e. only for class work.

Inclusion Policy – In this course, each voice in the classroom has something of value to contribute. Please take care to respect the different experiences, beliefs and values expressed by students and staff involved in this course. We support UMass Amherst's commitment to diversity, and welcome individuals of all ages, backgrounds, citizenships, disability, sex, education, ethnicities, family statuses, genders, gender identities, geographical locations, languages, military experience, political views, races, religions, sexual orientations, socioeconomic statuses, and work experiences.

Title IX Policy – UMass is committed to fostering a safe learning environment by responding promptly and effectively to complaints of all kinds of sexual misconduct. If you have been the victim of sexual violence, gender discrimination, or sexual harassment, the university can provide you with a variety of support resources and accommodations. If you experience or witness sexual misconduct and wish to report the incident, please contact the UMass Amherst Equal Opportunity (EO) Office (413-545-3464 | equalopportunity@admin.umass.edu) to request an intake meeting with EO staff. Members of the CICS community can also contact the CICS director of diversity and inclusive community development.

Academic Honesty Policy – Unless specifically stated in the assignment description, all work submitted in class must be your own. You may not copy portions or entire assignments from other students, pages on the internet, or generative AI engines such as ChatGPT. Reflections are graded on your thinking, not your prose, so a plain honest paragraph in your own words is worth more here than anything a chatbot writes for you. If issues arise, I will follow the policies outlined here: www.umass.edu/honesty

I reserve the right to make changes to this syllabus as needed to incorporate current events and to better support positive student outcomes.

Class Schedule (Subject to Change)

Monday sections meet Sep 14 – Dec 14; Wednesday sections meet Sep 9 – Dec 9. The two tracks cover the same material within a few days of each other, so the guest lectures fall on different weeks.

Monday (Sec. A, B)	Wednesday (Sec. C, D)	Topic
Sep 14	Sep 9	Two Symbols, Everything Else — where computers came from, why 0 and 1, and the logic gates that turn two symbols into everything else. Syllabus review and icebreakers.
Sep 21	Sep 16	Where the Bits Actually Live — memory as a numbered street of bytes, the cost of distance, and how an SSD works.
Sep 28	Sep 23	Counting With Two Fingers — invent a number system, then meet binary and hexadecimal.
Oct 5	Sep 30	How “a” Became a Number — design an alphabet code, then meet ASCII, Unicode, and why emoji break.
Oct 19	Oct 7	Guest lecture (speaker TBA)
—	Oct 14	Guest lecture (speaker TBA)
—	Oct 21	Guest lecture (speaker TBA)
Oct 26	Oct 28	Pictures Are Just Grids — pixels, colour as three numbers, and why an uncompressed photo is absurdly large.
Nov 2	Nov 4	Making It Smaller — compression, the hard limit on it, and what lossy formats quietly throw away.
Nov 9	Nov 18	Who Is Allowed to Read That? — the operating system as referee, and the permission bits that decide who may touch a file.
Nov 16	Tue Nov 24	Bits That Leave the Building — IP addresses, packets, DNS, and measuring the speed of light from a laptop.
Nov 23	—	Guest lecture (speaker TBA)
Nov 30	Dec 2	Guest lecture (speaker TBA)
Dec 7	—	Guest lecture (speaker TBA)
Dec 14	Dec 9	Wrap-Up and Final Show-and-Tell — student presentations, and where to go next.

No class: Indigenous Peoples’ Day (Mon, Oct 12), Veterans’ Day (Wed, Nov 11), Thanksgiving recess (Wed, Nov 25). **Wednesday sections:** the university follows a Wednesday schedule on **Tuesday, Nov 24** — we meet that day at our usual time.